



CYBERSECURITY LABORATORY REPORT

Building and Validation of a Cybersecurity Laboratory Environment
using Kali Linux, Docker, OWASP Juice Shop,
Nmap, Burp Suite and Wireshark



KALI LINUX
Attacker Machine



DOCKER
Container Platform



OWASP JUICE SHOP
Vulnerable Web
Application



NMAP
Network Scanning
& Enumeration



BURP SUITE
Web Traffic
Interception



WIRESHARK
Network Traffic
Analysis



SUBMITTED BY

NAKSHATHRA N

OBJECTIVE:

The objective of this task was to build a safe and isolated cybersecurity laboratory environment for practicing penetration testing, vulnerability assessment, web application testing, and network analysis. The lab environment consists of a Kali Linux attacker machine and OWASP Juice Shop as the vulnerable target application.

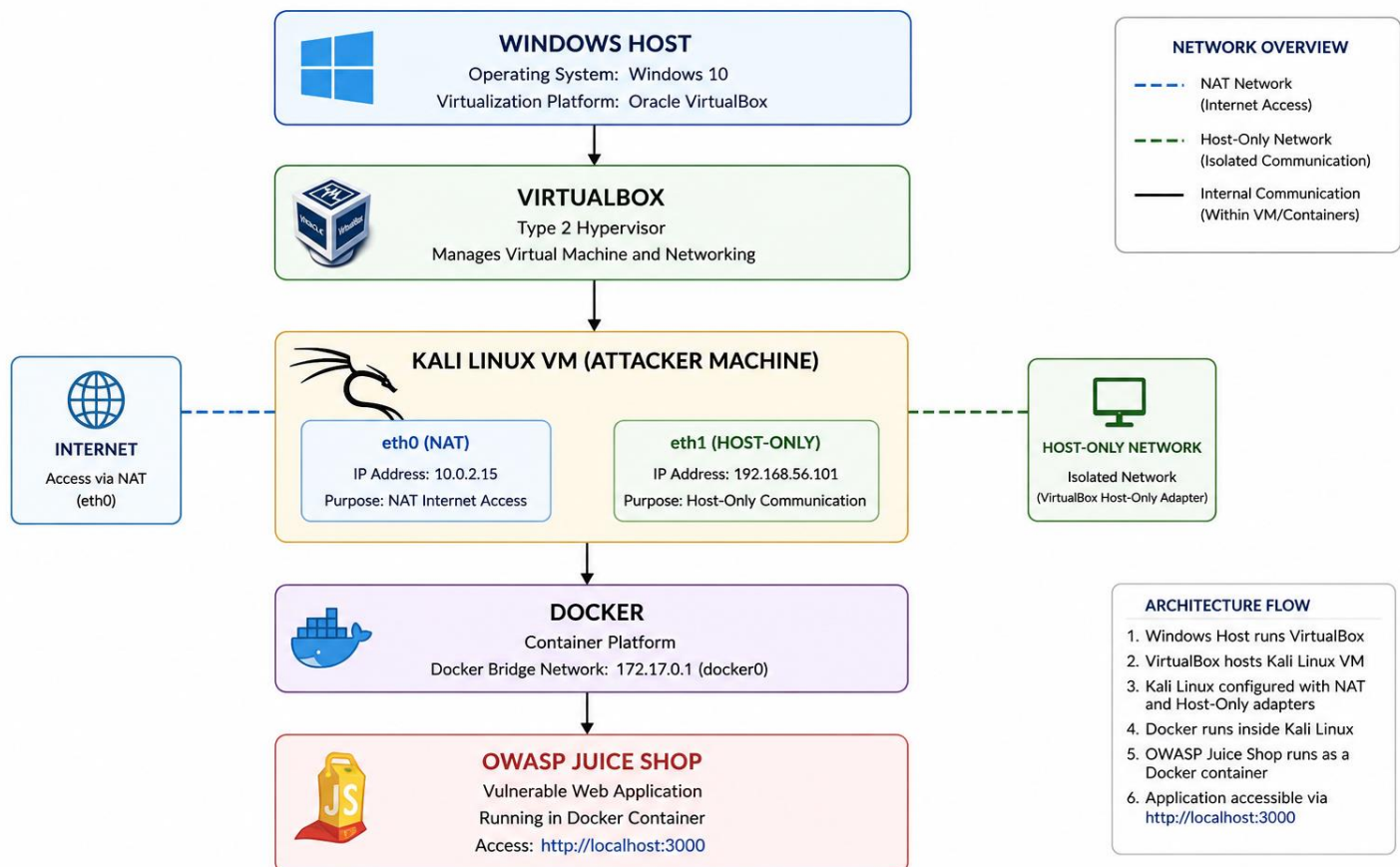
LAB ARCHITECTURE:

The lab environment was created using Oracle VirtualBox.

Components:

- Kali Linux (Attacker Machine)
- OWASP Juice Shop (Vulnerable Web Application)
- NAT Network Adapter
- Host-Only Network Adapter

LAB ARCHITECTURE DIAGRAM



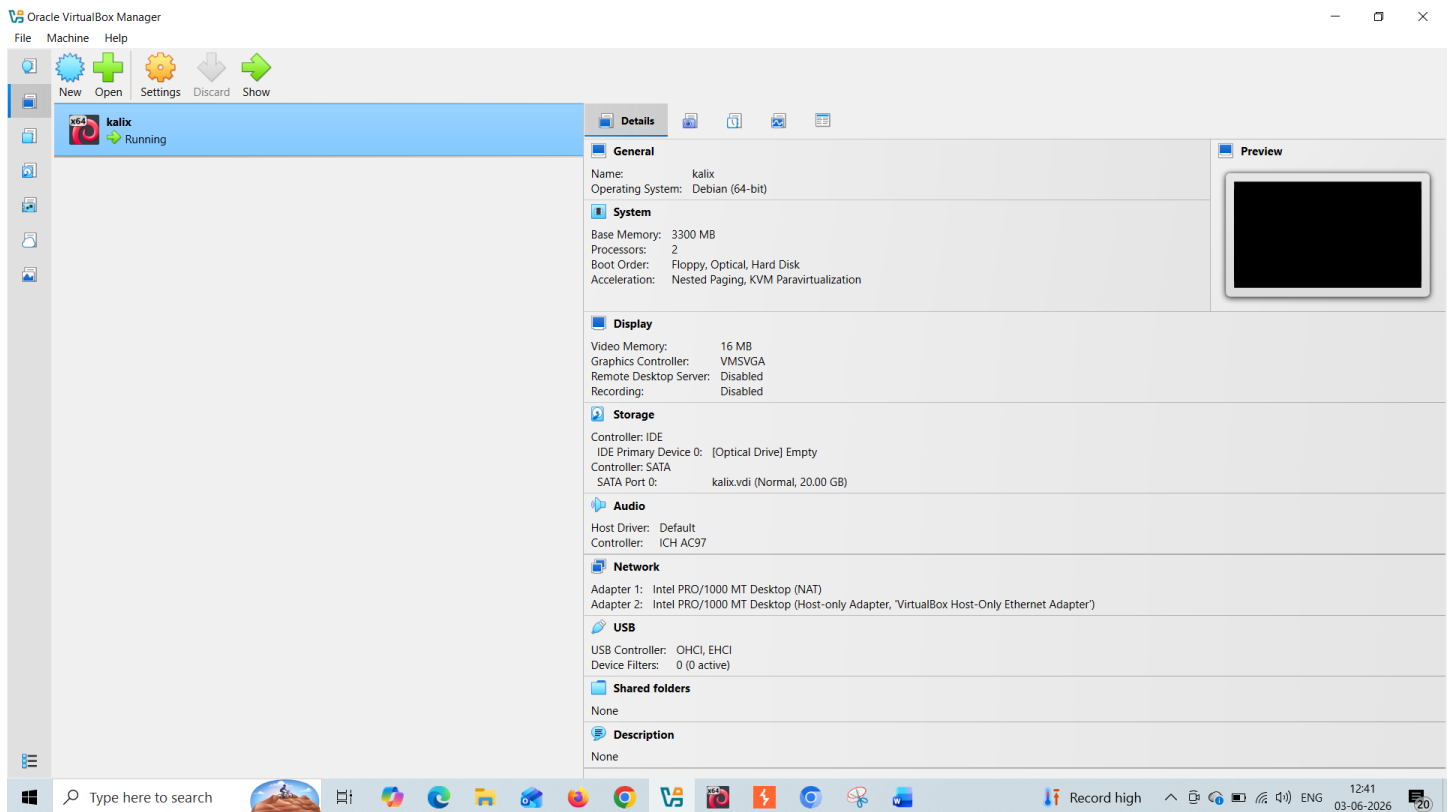


Figure 1: Oracle VirtualBox VM Configuration

SYSTEM REQUIREMENTS:

- Operating System: Windows 10
- RAM: 12 GB
- Virtualization: Enabled
- Virtualization Platform: Oracle VirtualBox
- Attacker Machine: Kali Linux
- Vulnerable Application: OWASP Juice Shop

VIRTUAL MACHINE CONFIGURATION:

Kali Linux was configured with:

- Adapter 1: NAT
- Adapter 2: Host-Only Adapter
- Internet connectivity enabled
- Host-only communication enabled

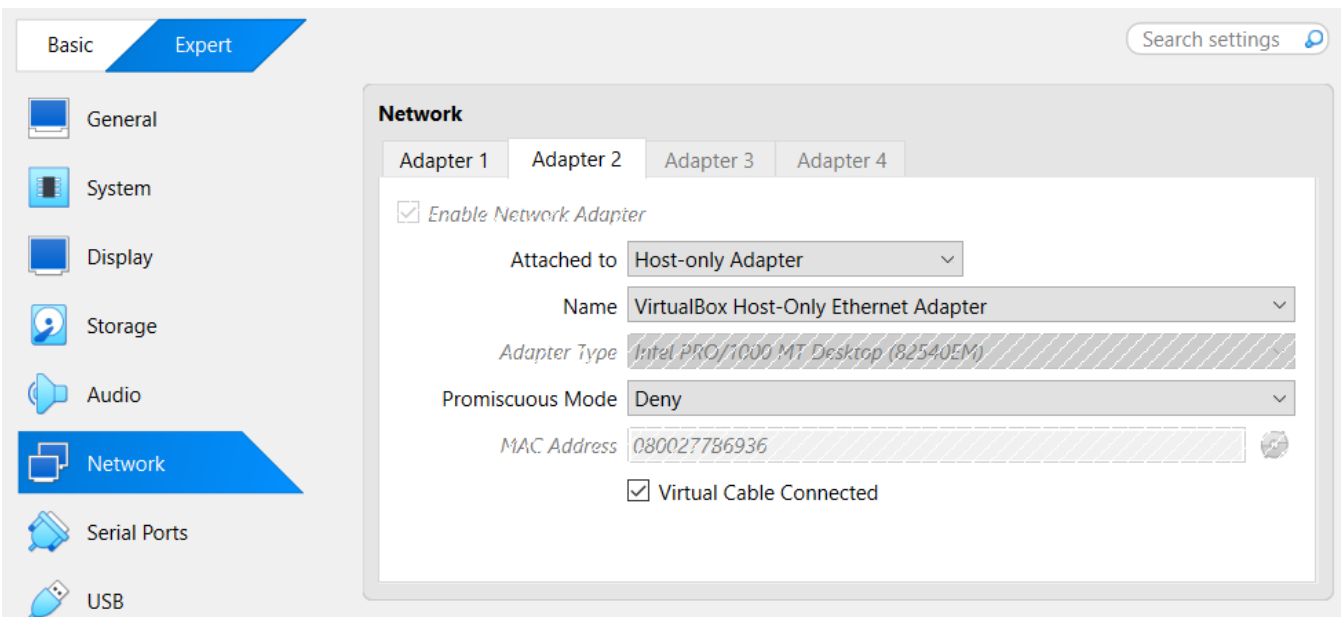
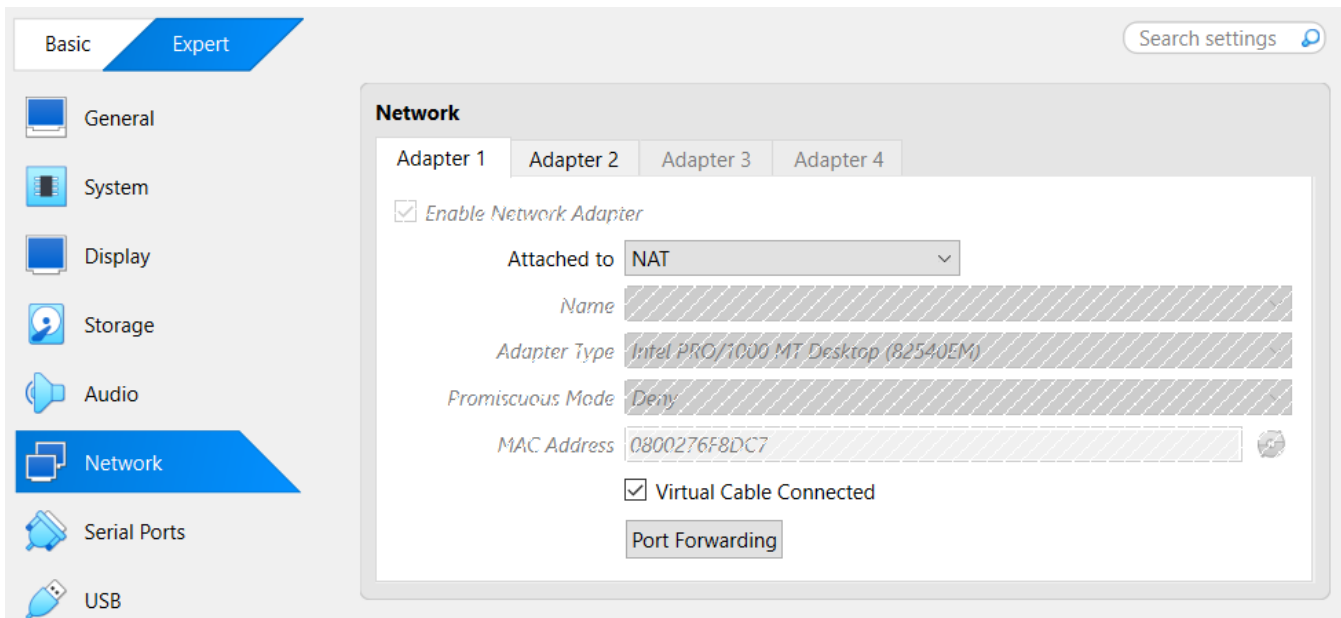


Figure 2: Virtual Network Adapter Configuration

NETWORK CONFIGURATION:

The network configuration was verified using the 'ip a' command.

Results:

- eth0 (NAT): 10.0.2.15
- eth1 (Host-Only): 192.168.56.101
- docker0 (Docker Bridge): 172.17.0.1

Table 1: IP Addressing Information

Interface	IP Address	Purpose
eth0	10.0.2.15	NAT Internet Access
eth1	192.168.56.101	Host-Only Communication
docker0	172.17.0.1	Docker Bridge Network

```
(naksha@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6f:8d:c7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86160sec preferred_lft 86160sec
    inet6 fd17:625c:f037:2:bf2c:4743:95b5:3bb3/64 scope global temporary dynamic
        valid_lft 86374sec preferred_lft 14374sec
    inet6 fd17:625c:f037:2:a00:27ff:fe6f:8dc7/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86374sec preferred_lft 14374sec
    inet6 fe80::a00:27ff:fe6f:8dc7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7b:69:36 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 595sec preferred_lft 595sec
    inet6 fe80::647:55b0:2186:e335/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether fa:11:68:3f:7a:71 brd ff:ff:ff:ff:ff:ff
```

Figure 3: IP Address Verification

CONNECTIVITY VALIDATION:

Internet connectivity was verified using ping commands.

Commands Used:

ping -c 4 8.8.8.8

ping -c 4 google.com

Results:

- Internet access confirmed
- DNS resolution confirmed


```

(naksha@kali)-[~]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=6.01 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=4.98 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=7.94 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=5.11 ms

— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 4.980/6.011/7.939/1.181 ms

(naksha@kali)-[~]
$ ping -c 4 google.com
PING google.com (142.251.221.206) 56(84) bytes of data.
64 bytes from pnmaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=1 ttl=64 time=5.69 ms
64 bytes from pnmaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=2 ttl=64 time=5.16 ms
64 bytes from pnmaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=3 ttl=64 time=5.08 ms
64 bytes from pnmaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=4 ttl=64 time=5.36 ms

— google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 5.080/5.320/5.686/0.234 ms

```

Figure 4: Connectivity Validation Using Ping

OWASP JUICE SHOP DEPLOYMENT:

Docker was used to deploy OWASP Juice Shop.

Commands Used:

```
sudo docker pull bkimminich/juice-shop
```

```
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

The application was successfully accessed through:

<http://localhost:3000>

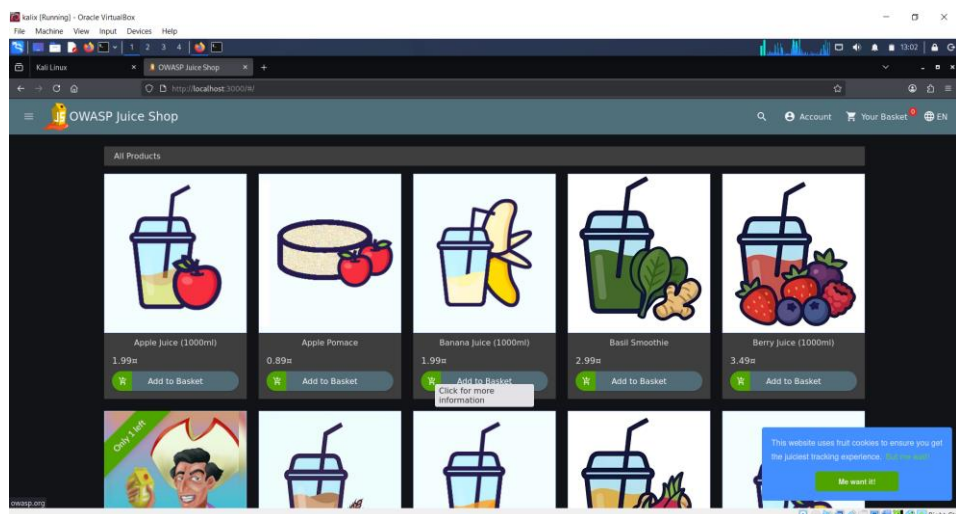


Figure 5: OWASP Juice Shop Deployment

NMAP VALIDATION:

Nmap (Network Mapper) is a network reconnaissance and security assessment tool used to discover hosts, identify open ports, and detect services running on a target system. It helps security professionals understand the network exposure of systems and verify service availability.

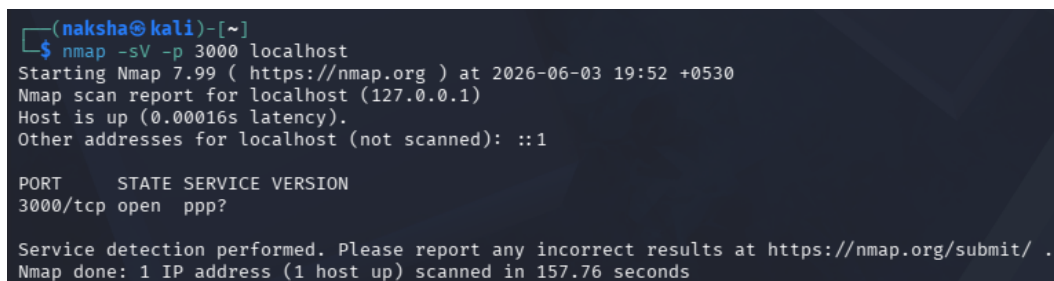
In this lab, Nmap was used to scan the local system and verify the availability of the OWASP Juice Shop application deployed through Docker. Service detection was performed on port 3000 to confirm that an active service was listening and accessible.

Command:

```
nmap -sV -p 3000 localhost
```

Results:

- Port 3000/tcp identified as open
- Service version detection confirmed an active service listening on port 3000
- The service was verified to be OWASP Juice Shop through successful browser access to <http://localhost:3000>



```
(naksha@kali)-[~]
$ nmap -sV -p 3000 localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-03 19:52 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
Other addresses for localhost (not scanned): ::1

PORT      STATE SERVICE VERSION
3000/tcp  open  ppp?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 157.76 seconds
```

Figure 6: Nmap Service Validation

BURPSUITE VALIDATION:

Burp Suite Community Edition is a web application security testing tool used to intercept, inspect, and analyze HTTP/HTTPS traffic between a web browser and a target application. It allows security professionals to observe web requests and responses, identify potential vulnerabilities, and understand application behavior.

In this lab, Burp Suite was configured as a proxy between the browser and the OWASP Juice Shop application. HTTP requests generated by the application were successfully intercepted and examined, confirming that traffic monitoring and analysis were functioning correctly.

Results:

- Proxy successfully configured
- HTTP requests intercepted
- HTTP traffic successfully analysed and Proxy functionality successfully validated

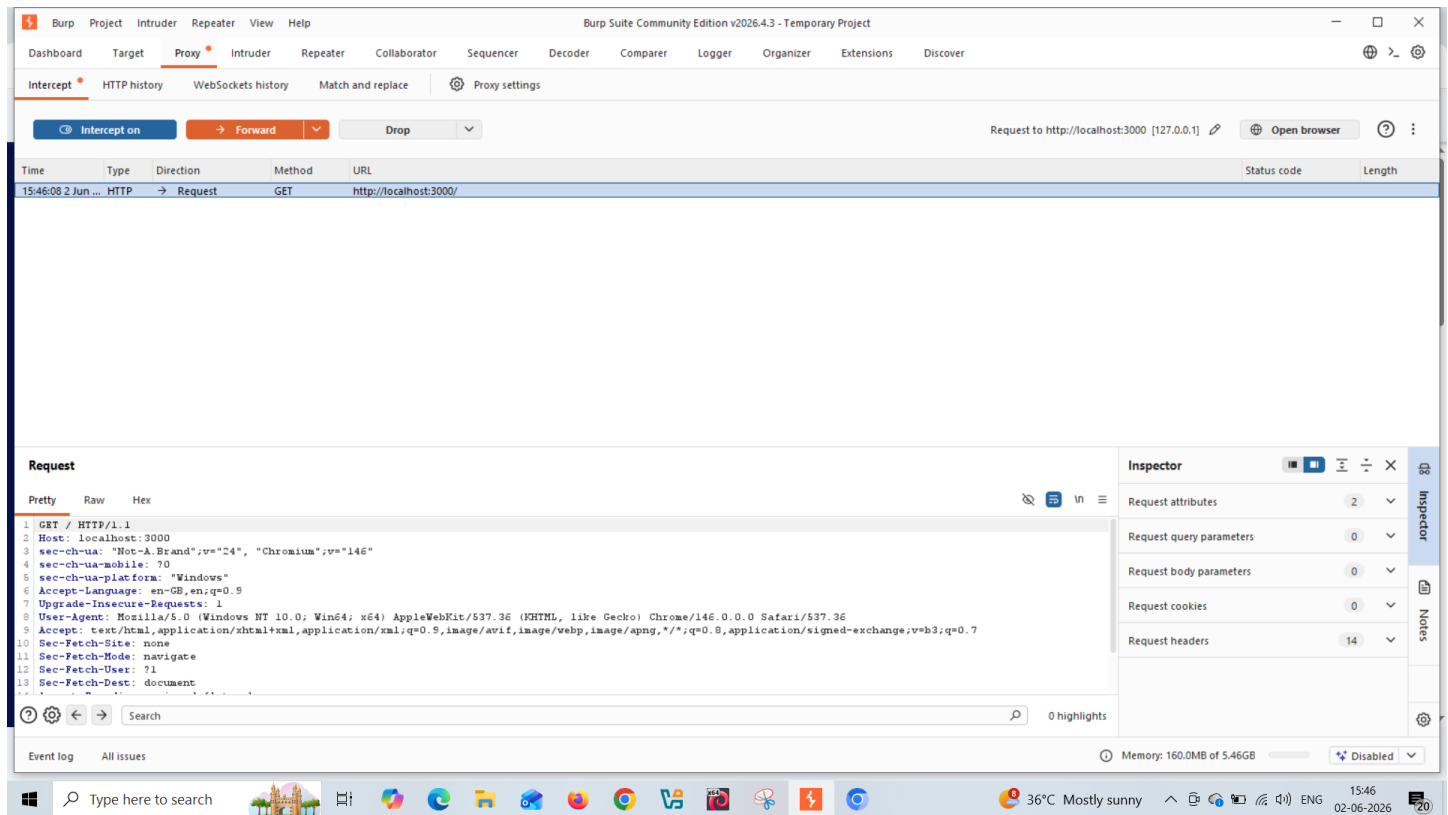


Figure 7: Burp Suite HTTP Interception

WIRESHARK VALIDATION:

Wireshark is a network protocol analyzer used to capture and inspect network traffic in real time. It allows cybersecurity professionals to monitor communication between devices, analyze network protocols, troubleshoot connectivity issues, and investigate security-related events.

In this lab, Wireshark was used to capture and analyze ICMP traffic generated through ping requests, validating successful network communication and packet analysis capabilities.

Result:

- Network interfaces successfully detected
- Live network traffic captured and analysed
- ICMP request and reply packets identified
- Packet analysis environment successfully validated

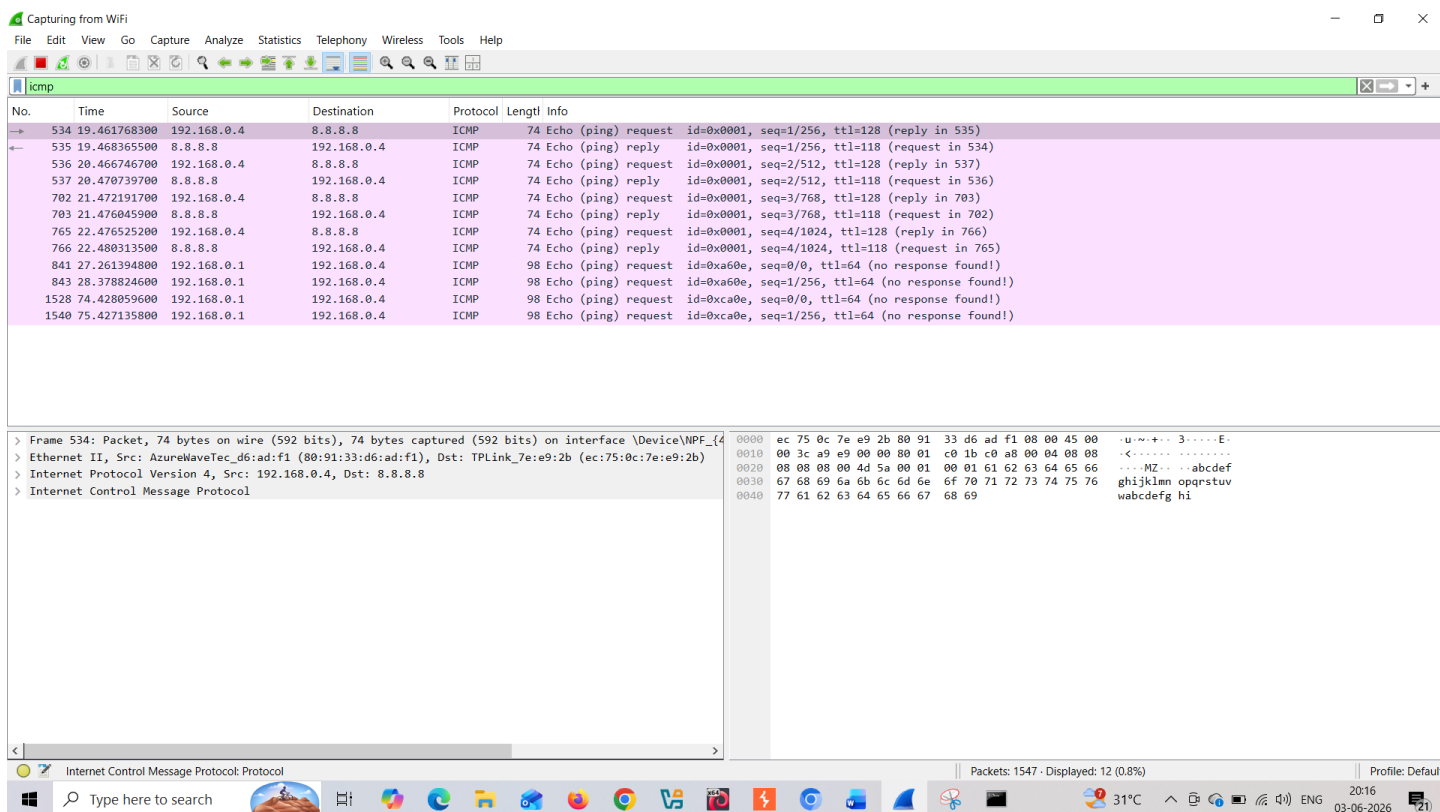


Figure 8: Wireshark Packet Capture

KEY OUTCOMES:

- Successfully deployed Kali Linux in Oracle VirtualBox
- Configured NAT and Host-Only network adapters
- Verified network connectivity and DNS resolution
- Successfully deployed OWASP Juice Shop using Docker
- Validated application accessibility through port 3000
- Performed service enumeration using Nmap
- Successfully intercepted HTTP traffic using Burp Suite
- Captured and analyzed network traffic using Wireshark
- Established a functional cybersecurity laboratory environment

CONCLUSION:

The cybersecurity laboratory environment was successfully designed, deployed, and validated using Oracle VirtualBox, Kali Linux, Docker, and OWASP Juice Shop. Network connectivity and service availability were verified through multiple validation techniques including Ping, Nmap, Burp Suite, and Wireshark. The completed environment provides a secure and isolated platform for cybersecurity training, web application security testing, vulnerability assessment, and network traffic analysis.

REFERENCES:

1. OWASP Juice Shop Documentation - <https://owasp.org/www-project-juice-shop/>
2. Kali Linux Documentation - <https://www.kali.org/docs/>
3. Docker Documentation - <https://docs.docker.com/>
4. Nmap Documentation - <https://nmap.org/docs.html>
5. PortSwigger Burp Suite Documentation - <https://portswigger.net/burp/documentation>
6. Wireshark Documentation - <https://www.wireshark.org/docs/>